Parisa Dehghan

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Professional Summary

Research and Development and IoT Specialist. Highly motivated and detail-oriented M.Sc. candidate in Particle Physics and Field Theory with a strong foundation in data analysis, machine learning, and quantum field theory. Proficient in Python, C++, and SQL, with hands-on experience in numerical simulations, high-performance computing, and scientific machine learning. Skilled in designing and implementing algorithms, deploying machine learning models on low-power devices, and conducting research in complex systems. Passionate about applying computational techniques to solve real-world problems in physics and beyond.

Work Experience

Internet of Things Specialist, Iran IoT Expansion Center — Since May 2025

1. Smart Health-Monitoring Web Application

- End-to-end architecture: Designed a real-time IoT monitoring system that ingests telemetry from smart wristbands (GPS, blood-pressure, SpO_2 and fall-detection).
- Device simulation: Emulated LoRa-enabled wearables in Node-RED, producing realistic, high-frequency sensor streams that are published via MQTT to the backend.
- Data pipeline: Built a Node.js REST/MQTT service that validates, timestamps, and persists the telemetry into a local SQL database (SQLite/MySQL).
- Developed a responsive web UI using HTML/CSS and JavaScript.

2. IoT Instructor

- Delivered a 6-week “Intro to IoT” online course for 8–12-year-old students.

Research and Development Specialist, Intelligent Micronano Sensors (Lotus IMNS) — Since May 2024

1. AFM Sensor Development

- Analyzed fabrication data in Python and conducted multiphysics simulations in COMSOL to optimize performance and structured datasets to support machine learning–driven process improvements.
- Designed and fabricated AFM sensors on silicon wafers using UV lithography, RIE, and wet etching.
- Deposited gold thin films via PVD for functional surface enhancement.
- Engineered high-aspect-ratio tips through iterative process optimization, improving tip sharpness and release yield.
- Performed structural and surface characterization using SEM, AFM, and optical microscopy.
- Created mask designs using CorelDraw.

2. Microneedle Patch Development

- Fabricated PDMS molds from silicon microneedle masters using microfabrication techniques.
- Optimized hydroxyapatite (HA) loading via centrifugation and engineered PVP formulations for transparent, mechanically robust patches.
- Conducted formulation R&D, improving flexibility and reducing brittleness through iterative testing.
- Created centrifuge part designs using Catia.

Education

M.Sc. in Particle Physics and Field Theory, Shahid Beheshti University, Tehran, Iran 2021 – 2024

- **GPA:** 14.13/20
- **Relevant Courses:** Quantum Field Theory II (18.23/20), Geometry and Topology (15/20), Gravity I (17/20), Advanced Quantum Mechanics (16.02/20), Advanced Statistical Mechanics (15.9/20)

B.Sc. in Physics, Alzahra University, Tehran, Iran 2017 – 2021

- **GPA:** 15.66/20
- **Relevant Courses:** Numerical Calculations (4/4), Stochastic Processes (4/4), Advanced Quantum Mechanics I (4/4), Computer Programming (Theoretical & Practical) (4/4)

Diploma in Mathematics and Physics, Motahare High School, Tehran, Iran 2013 – 2016

- **GPA:** 19.38/20 (4/4)

Certifications

- **SANS SEC505 (LIAN group, Sep 2025).**
- **Network+ (LIAN group, Sep 2025).**
- **Security+ (LIAN group, Sep 2025).**
- **Certified Ethical Hacker (CEH) (Maktabkhooneh Jadi's course, July 2025).**
- **Software test bootcamp (Behpardaz Hamrah Samaneh Aval Behsa, May 2024):** Data analysis and software test using SQL server and Oracle.
- **DESY Remote Summer School 2023:** Data analysis, simulations, machine learning, and high-performance computing using Python, C++, and Fortran.
- **Greedy Algorithms, Minimum Spanning Trees, and Dynamic Programming (Stanford University, Coursera, 2023).**
- **Workshop on Stochastic Thermodynamics (ICTP, May 2023).**
- **Applied Machine Learning in Python (University of Michigan, Coursera, October 2022).**
- **Workshop on Topological Data Analysis (Shahid Beheshti University, August 2022).**
- **Python Course (Alzahra University, October 2021).**
- **Introduction to C ,Introduction to C# and Intermediate SQL (SoloLearn).**

Skills

Programming & Tools:

- Proficient in Python (TensorFlow, Keras, OpenCV, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn)
- Strong C/C++ programming skills for algorithm design and high-performance computing
- Experience with MATLAB for computational work and STAR-CCM+ for fluid mechanic simulation.
- Database management: SQL Server, Oracle
- Web development: API design for text classification
- Version control: Git/GitHub

Machine Learning & Data Analysis:

- Supervised and unsupervised learning (classification, regression, clustering)
- Neural networks (RNN, LSTM, autoencoders, GCN)
- Topological Data Analysis (TDA) using GUDHI, Dionysus, JavaPlex
- Data preprocessing, feature engineering, and visualization

Scientific Computing & Simulations:

- Numerical simulations of quantum field theories

- High-performance computing with C++ and Fortran
- Simulation tools: COMSOL, Rivet (Docker)
- Monte Carlo methods and stochastic differential equations

Hardware & Deployment:

- Raspberry Pi setup and remote programming via SSH
- TinyML: Model optimization (quantization, pruning) for low-power devices
- Sensor data analysis and deployment

Design & Creative Tools:

- 3D modeling: 3ds Max (jewelry design), CATIA
- Graphic design: Adobe InDesign (brochures, reports), Adobe Illustrator (logos)
- Photography and image editing: Adobe Lightroom, Photoshop

Languages:

- Persian (Native)
- English (Fluent, TOEFL iBT 93)
- French (Intermediate, Duolingo)

Projects

Environmental Monitoring

- Deployed a real-time sensor data analysis model on a Raspberry Pi, achieving high energy efficiency.

Bubble Rise Simulation (SPH & STAR-CCM+)

- Developed a Python SPH model and an automated STAR-CCM+ macro to simulate and analyze micron-sized bubble rise dynamics, including scaled CFD modeling, theoretical comparison, and automated data export.

Phase Transition in the Ising Model

- Implemented the Metropolis Monte Carlo algorithm with scrambling to determine phase transitions.

Graph Convolutional Networks

- Authored a research paper on GCNs and their applications in complex systems.

Stock Price Prediction

- Developed an RNN model with stacked LSTM and Bidirectional LSTM for accurate stock price forecasting.

Anomaly Detection

- Designed an autoencoder neural network to detect fraudulent financial transactions.

Sentiment Analysis

- Analyzed Snapfood reviews using a Naive Bayes classifier.

Cryptocurrency Classification

- Applied KMeans clustering to classify cryptocurrencies and analyze market trends.

Breast Cancer Detection

- Used unsupervised machine learning for early detection of breast cancer.

Particle Accelerator Simulation

- Simulated high-energy processes using the Rivet container in Docker.

Knapsack Problem

- Designed a dynamic programming solution for resource allocation optimization.

Huffman Coding

- Developed a data compression algorithm, reducing file sizes by up to 30%.

Minimum Spanning Tree

- Applied Kruskal's algorithm to optimize network design efficiency by 20%.

Hamshahri Newspaper Dataset

- Conducted comprehensive data processing, visualization, and feature engineering.

MRI Image Analysis

- Identified structural brain disorder differences using advanced image processing techniques.

Topological Data Analysis

- Applied persistent homology and Mapper algorithms to biological and network data.

Stochastic Thermodynamics

- Conducted numerical simulations of stochastic differential equations for molecular motors and nanoscale engines.

Calculator Application

- Built a basic calculator to practice control structures and methods.

To-Do List Application

- Developed a task management tool using arrays and file handling.

Library Management System

- Designed an OOP-based system showcasing class design and inheritance.

Image Classification

- Optimized a neural network using quantization and pruning, reducing model size by 50% while maintaining >90% accuracy.

Hobbies

- Ballet
- Painting
- Guitar
- Analog photography